

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q18-1048

Longview2 345 kV  
August 20, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1048. The Interconnection Customer proposes a 1275.1 MW natural gas combined cycle facility interconnecting at the Longview2 345 kV substation in the West Virginia area. The request was received on August 28, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 431 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$304,316,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                     | Contingency                                     | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------|-------------------------------------------------|-------------|--------------|----------|
| Declo to Orchard Substation 138 kV L   | Crandall to Aspen 345 kV                        | 132.7       | 361          | 5.8      |
| Villa Mar 2 1658 to Orchard 138 kV L   | Crandall to Aspen 345 kV                        | 130.9       | 375          | 5.8      |
| Villa Mar 2 1658 to Aspen 138 kV Line  | Crandall to Aspen Substation 345 kV             | 129.0       | 392          | 5.8      |
| Weyers Cave to Grottoes 138 kV L       | Crandall to Dayton 345 kV                       | 122.6       | 533          | 6.8      |
| Dayton to Harrisonburg 138 kV line     | Chatom to Dayton 345 kV                         | 122.5       | 610          | 9.0      |
| Dayton 345/138kV XFMR                  | Dayton to Chatom 345 kV L                       | 120.7       | 644          | 9.0      |
| Chatom 345/138 kV TX                   | Dayton 345 kV/138 kV TX                         | 120.2       | 637          | 9.1      |
| Lowmoor to Aspen 345 kV                | Bath County to Longview2 345 kV L               | 117.5       | 1,660        | 39.9     |
| Redland to Carodean 138 kV             | Dayton to Crandall 345 kV line                  | 115.8       | 443          | 6.6      |
| Crandall Substation to Carodean 138 kV | Dayton to Crandall 345 kV                       | 113.9       | 475          | 6.6      |
| Clover to Bear 345 kV L                | Mecklenburg Power Station to Person 345 kV Line | 113.1       | 3,332        | 8.3      |
| Weyers Cave to Greenbank 138 kV        | Crandall to Dayton 345 kV                       | 112.7       | 476          | 6.8      |
| Aspen 345/138kV Transformer            | Crandall to Aspen 345 kV                        | 112.0       | 703          | 13.8     |
| Bath County to Longview2 345 kV Line   | Aspen to Lowmoor 345 kV                         | 111.9       | 1,332        | 47.2     |
| Crandall 345/138 kV XFMR               | Crandall to Dayton 345 kV L                     | 111.7       | 535          | 6.6      |
| Crandall to Aspen 345 kV               | Bath County to Longview2 345 kV                 | 111.7       | 1,316        | 23.0     |

| Monitored Facility                                     | Contingency                   | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------------|-------------------------------|-------------|--------------|----------|
| Person to Mecklenburg Power Sta Substation 345 kV Line | Clover to Bear 345 kV         | 110.3       | 3,204        | 8.0      |
| Bear to Leavitt 345 kV Ckt 1                           | Leavitt to Bear 345 kV Line 2 | 103.0       | 2,239        | 5.6      |
| Santee Cooper to Mountaineer Wind Energy Ctr 345 kV    | Spindle 500 kV/345 kV TX      | 102.7       | 1,250        | 9.1      |

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                              | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------|---------------------------------------------|---------------|-----------------------|
| Declo to Orchard 138 kV               | Rebuild line with higher-capacity conductor | \$8,862,000   | 37% (\$3,286,000)     |
| Villa Mar 2 1658 to Orchard 138 kV L  | Rebuild line with higher-capacity conductor | \$5,713,000   | 37% (\$2,119,000)     |
| Aspen to Villa Mar 2 1658 138 kV      | Rebuild line with higher-capacity conductor | \$2,524,000   | 37% (\$936,000)       |
| Grottoes to Weyers Cave 138 kV        | Rebuild line with higher-capacity conductor | \$7,716,000   | 34% (\$2,612,000)     |
| Dayton to Harrisonburg 138 kV line    | Rebuild line with higher-capacity conductor | \$1,926,000   | 37% (\$707,000)       |
| Dayton 345/138kV XFMR                 | Replace transformer with higher-rated unit  | \$11,589,000  | 37% (\$4,254,000)     |
| Chatom 345/138 kV Auto                | Replace transformer with higher-rated unit  | \$14,979,000  | 33% (\$5,005,000)     |
| Lowmoor to Aspen 345 kV               | Rebuild line with higher-capacity conductor | \$63,504,000  | 34% (\$21,350,000)    |
| Redland Substation to Carodean 138 kV | Rebuild line with higher-capacity conductor | \$1,344,000   | 38% (\$510,000)       |
| Crandall to Carodean 138 kV           | Rebuild line with higher-capacity conductor | \$3,989,000   | 38% (\$1,514,000)     |
| Bear to Clover 345 kV                 | Rebuild line with higher-capacity conductor | \$108,771,000 | 8% (\$8,545,000)      |
| Weyers Cave to Greenbank 138 kV       | Rebuild line with higher-capacity conductor | \$4,015,000   | 34% (\$1,371,000)     |
| Aspen 345/138kV Transformer           | Replace transformer with higher-rated unit  | \$19,336,000  | 22% (\$4,288,000)     |
| Bath County to Longview2 345 kV Line  | Rebuild line with higher-capacity conductor | \$325,458,000 | 50% (\$162,698,000)   |
| Crandall 345/138 kV XFMR              | Replace transformer with higher-rated unit  | \$21,821,000  | 40% (\$8,681,000)     |
| Crandall to Aspen 345 kV              | Rebuild line with higher-capacity conductor | \$141,001,000 | 36% (\$50,746,000)    |

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Mecklenburg Power Sta to Person 345 kV Line         | Rebuild line with higher-capacity conductor | \$114,178,000 | 9% (\$10,551,000)     |
| Bear to Leavitt 345 kV Ckt 1                        | Reconductor line                            | \$69,786,000  | 8% (\$5,596,000)      |
| Santee Cooper to Mountaineer Wind Energy Ctr 345 kV | Reconductor line                            | \$15,443,000  | 62% (\$9,547,000)     |
|                                                     | Total allocated to project                  |               | \$304,316,000         |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.